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ORIGINAL ARTICLE

A comprehensive strategy for identifying plagiarism in academic submissions

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Abstract

Plagiarism is a major problem in education, especially in higher education environments. To address this problem, a comprehensive detection method is proposed, utilizing cutting-edge models like Bidirectional Encoder Representations from Transformers (BERT) and cosine similarity for detecting direct copy and semantic changes. The methodology involves evaluating the BERT model and cosine similarity first, then performing pairwise comparisons to determine how similar a student's work is to a reference text. If the semantic changes are higher, the content is classified as paraphrased; otherwise, it is considered direct copy in cases where the similarity threshold is greater than 30%. When the similarity threshold is less than or equal to 30%, the method moves to online plagiarism detection. This includes web scraping, generating search queries, and employing Term Frequency-Inverse Document Frequency (TF-IDF) vectorization with cosine similarity to assess the similarities between students' content and online content. As a result, the study detects online plagiarism and distinguishes between plagiarizing and authentic content. The purpose is to provide an effective method to maintain academic integrity and enhance originality in student work. In similarity testing on MIT plagiarism detection datasets, this study demonstrated an average recall of 80%, precision of 68%, accuracy of 71%, and f1-score of 74%. With a threshold of 0.2, the average precision for online plagiarism detection reached 65%, accompanied by a 56% recall, an accuracy of 60%, and an f1-score of 60%. The results of the study demonstrate the efficiency of the strategy in enhancing plagiarism detection accuracy while reducing false positives.

Keywords Cosine similarity · Bidirectional encoder representations from transformers · Plagiarism · Paraphrasing · Web scraping · Term frequency-inverse document frequency

1 Introduction

The internet has made knowledge easily accessible in recent decades. Plagiarism [1], a serious problem in education and the arts, is brought on by this accessibility. Plagiarism occurs when an individual employs another person's words, ideas, or creative work without properly acknowledging the source. Plagiarism is committed by dishonest people in a variety of ways. Direct copy (word-for-word in text) [2] is the most popular form of plagiarism, involving directly copying text from one or more sources, occasionally with minor adjustments. Another method is the paraphrasing [3] approach, which modifies the text's grammar by substituting synonyms for words. The sentences have been rearranged, and some of the original language may have been left out. The borrowing of ideas is a more complicated form of plagiarism. It might use a variety of different techniques in

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addition to basic text changes, making it difficult to identify. Lastly, a common practice involves replicating content from online sources without appropriately crediting the original sources. Although these acts are strictly forbidden, they are frequently motivated by a desire to improve academic performance. In addition to compromising the integrity of the educational process, such practices raise ethical concerns for both educators and learners.

Therefore, these unethical practices not only prevent the development of essential skills, such as critical thinking, but also prevent students from achieving a comprehensive understanding of the concept. As a result, plagiarism affects the true performance of an individual's abilities, resulting in an instance where academic achievement fails to represent genuine knowledge and effort. This affects the equity of the assessment system, making it harder to provide helpful feedback and measure real academic achievement. So, it is essential to identify plagiarism and take steps to reduce it to maintain the integrity of education. However, dealing with plagiarism might be challenging. Although there are programs like Turnitin [4], CopyCatch [5] and others that are intended to detect plagiarism, they can be costly for some educational institutions, including schools and colleges. As a result, they may not always be accessible, which can pose challenges in maintaining academic integrity. There have been several methods for detecting plagiarism despite this. Steve et al. [6] presented an automated approach that employs neural network methods and feature-based plagiarism detection, providing a clear viewpoint on relevance assessment. The development of a scheduled algorithm by Alan et al. [7] made a substantial contribution to the maintenance of academic integrity. A closer evaluation of these two methods reveals that they can detect plagiarism that results from a direct copy of certain resources. However, they fail in detecting plagiarism that arises from paraphrasing and the use of online content. Therefore, a method is essential that is free of cost and can identify plagiarism from paraphrasing, copying from classmates, and using online content without proper credit.

This study aims to provide a cost-effective plagiarism detection technique for educational institutions where commercial plagiarism detection tools are unavailable due to financial constraints. This study also addresses a significant gap in current systems by focusing on the detection of plagiarism within student cohorts, particularly when students paraphrase their peers' work line by line in an attempt to avoid detection. Furthermore, the study is designed to identify instances of plagiarism that involve content copied directly from online sources without paraphrasing. By offering an accessible and effective solution, this research seeks to promote academic integrity in schools, colleges, and universities, regardless of their access to commercial plagiarism detection software.

This proposed methodology aims to improve current approaches by addressing their limitations and combining several proven techniques. This approach uses Term Frequency-Inverse Document Frequency (TF-IDF) vectorization [8] and cosine similarity [9] to calculate the percentage of direct copying, taking advantage of TF-IDF's ability to give numerical values to words according to their relevance in a document. This will allow us to quantify the similarity of documents using cosine similarity, a conventional but efficient method for finding exact text matches. But since paraphrasing is a common part of academic writing, this suggested method incorporates Bidirectional Encoder Representations from Transformers (BERT) [10], an effective deep learning (DL) [11–13] model renowned for its ability to interpret real language. DL models such as BERT's proficiency in identifying subtle semantic connections between sentences improve the system's capacity to identify paraphrases. DL models have proven effective in various domains, as demonstrated in studies such as Kumar et al. [12] and Kaur et al. [13]. This method will compute the maximum result of the BERT model and cosine similarities in order to find similarities. To identify internet information quickly and accurately for online plagiarism detection, TF-IDF and cosine similarity are combined with the "Google search" package [14]. This all-encompassing strategy seeks to successfully address the difficulties presented by different types of plagiarism.

Several key contributions are the focus of this comprehensive approach:

- To measure the degree of similarity between the texts of both original and plagiarized content.
- To differentiate between documents that have been fully copied and those that have been paraphrased.
- To improve the capability of identifying contents that have been copied directly from online sources.

The result is given as a percentage, which offers a clear indicator of textual similarity for in-depth examination. By fulfilling these goals, this study aims to improve plagiarism detection techniques and support the more general objective of maintaining academic integrity.

The rest of the paper is organized in this manner: Sect. 2 describes the relevant works completed. Section 3 provides a brief overview of the experimental setup. Section 4 presents the results and discussion. Section 5 provides the conclusion and recommendations for future developments.

2 Related work

Several papers have been written about detecting plagiarism. This section describes some recent works that are detectable for plagiarism.

The Rabin-Karp technique was used by Siahaan et al. in the paper [15] in question to hash documents and compare them effectively. Data preprocessing techniques used in this method include tokenization, stemming, and stop word removal. After that, hashing is used to calculate word values while utilizing huge prime numbers and taking into account N-grams. The program determines similarities and computes percentages of plagiarism by comparing the hash values. The Rabin-Karp method performs similarity detection quickly and efficiently. Even though the Rabin-Karp approach is quite effective at finding similarities quickly, there are various things to be aware of. This method, which is mostly intended for local string matching, has trouble distinguishing between information that has been taken from the internet or paraphrased. When a text is heavily paraphrased or comes from an internet source, the Rabin-Karp technique's effectiveness could be limited.

Several methods have been investigated in the field of plagiarism detection for locating textual similarities between papers. Notably, the effectiveness of the Jaro-Winkler Distance algorithm and the Rabin-Karp algorithm in this field has been described by Leonardo et al. in the paper [16]. The Rabin-Karp technique, known for its ability to find instances of substrings in longer texts, has proven useful in determining the percentage of equivalent word counts between documents. The Jaro-Winkler Distance technique, on the other hand, focuses on analyzing the similarity of documents that are identical or extremely similar. These algorithms could have trouble distinguishing between subtle paraphrases and when there is a significant rearrangement of the text. Furthermore, because online content is always changing, its efficacy may be limited in situations involving content obtained from the internet. However, the proposed method enhances these methods by including TF-IDF and BERT embeddings to capture structural and semantic similarities. While Jaro-Winkler and Rabin-Karp are appropriate for exact matches, the suggested method improves their capabilities to detect contextual overlaps and subtle paraphrasing, overcoming limitations of existing string-based algorithms.

Sutoyo et al. [17] examined the two document similarity detection algorithms, Winnowing and Karp-Rabin. Preprocessing, which includes data parsing, stop word removal, punctuation, stemming, and word grouping (n-grams), is the first step of both methods. Then, hashing is used to calculate the hash values. These hash values produce a unique pattern for matching data and possible documents in the Rabin-Karp technique. The original text's k-grams are hashed using the rolling hash method. The Winnowing algorithm, in contrast, creates a distinct pattern by choosing the window's lowest hash values. The basic details regarding the algorithms and their processes are covered in this condensed edition. Though useful in many cases, these algorithms might not work well with heavily paraphrased language or in cases where there has been quite a bit of textual rearrangement. Furthermore, in some situations, accuracy may be impacted by their sensitivity to word order.

Novak et al. [18] add to the academic conversation by providing a thorough analysis of significant studies in the field of source-code plagiarism detection in academic settings. It establishes the framework for upcoming developments in this crucial area by highlighting current discoveries and research needs.

Jiffriya et al. [19] detect academic misconduct that encompasses the field of plagiarism detection. In this paper, existing methods for detecting plagiarism are thoroughly examined, with a focus on approaches that make use of similarity metrics and the vector space model. An important finding of the analysis was that, despite being somewhat more time-consuming, the trigram-based approach emerged as a viable option for improving the accuracy of plagiarism detection in textual materials. This method, which is based on the vector space model, gives priority to finding uncommon or infrequently occurring terms in the dataset, enhancing its ability to spot plagiarism. Furthermore, the comparison study proved the superiority of the cosine similarity metric over other metrics. Even though the trigram-based strategy is good at increasing the accuracy of plagiarism detection, it is important to remember that this method can be a little more time-consuming, which may affect real-time detection in some situations. Furthermore, although the trigram-based method prioritizes finding rare or unusual terms, it may have trouble correctly detecting paraphrased text that makes extensive use of common language.

Researchers previously investigated the use of the Vector Space Model (VSM) in plagiarism detection. In this experiment, textual paragraphs originating from query and collection vectors were ranked using the cosine similarity method. In order to estimate the ideal number of words to extract from each paragraph, Saptono et al. [20] also used conditional probability. By using threshold values of 0.3 for conditional probability and 0.2 for cosine similarity, the results showed outstanding performance, reaching an average precision rate of 54.28% and a perfect average recall rate of 100%. For the particular dataset and experiment circumstances, the threshold values of 0.2 for cosine similarity and 0.3 for conditional probability may have been the best ones. It's possible that these principles won't hold for all

datasets, topics, or writing styles. The effectiveness of plagiarism detection greatly depends on the properties of the text being examined.

Resta et al. [21] developed a novel strategy using data mining techniques to find similarities between students' final scientific works in order to solve the problem of plagiarism prevention. This method entailed determining the degree of similarity between an abstract and title of a student's final scientific paper using the cosine similarity methodology, preprocessing methods, and TF-IDF (Term Frequency-Inverse Document Frequency). The results were then compared to the database of completed projects, allowing the scientific effort to be approved or rejected based on a predetermined threshold value. However, the emphasis on comparing scientific publications according to their abstract and title may make it difficult to identify subtle cases of plagiarism that occur within the text itself. Analyzing abstracts and titles alone may not completely catch the subtler content modifications that constitute plagiarism, such as paraphrasing or incorporating ideas within the body of the text.

Moravvej et al. [22] provide a novel method for detecting plagiarism that incorporates attention-based LSTM networks, BERT word embeddings, and an improved Differential Evolution (DE) algorithm for weight initialization. They incorporate attention mechanisms into LSTMs, employ BERT for word embeddings, and use DE for weight initialization. This method detects plagiarism better than other methods on datasets like SNLI, MSRP, and SemEval2014. BERT embeddings and attention-based LSTM networks excel in capturing semantic context and sequential dependencies, making them effective for paraphrasing detection. However, DE algorithms and BERT limit their handling of structural overlaps and scalability. Computational complexity is introduced by the use of sophisticated methods like BERT embeddings and attention procedures. This complexity may limit these algorithms' real-time usefulness in scenarios where prompt and effective plagiarism detection is essential. This proposed method addresses this by integrating TF-IDF and cosine similarity for structural analysis with BERT for semantic similarity, ensuring a scalable and comprehensive plagiarism detection system.

The NLP research area is addressed with attention processes and sentence transformers to detect plagiarism, incorporating recent developments. In comparison to conventional methods, the proposed strategy increases the accuracy of plagiarism detection by utilizing cutting-edge deep learning techniques. This method uses attention mechanisms to understand and process text in context, instead of relying on slow and easily skipped word search and matching techniques. Rosu et al. [23] investigate three distinct methodologies in order to guarantee robustness and relevance. The superiority of systems using the BERT pre-trained model over GloVe and RoBERTa-based strategies is demonstrated by experimental data.

An online detection system was created to combat the abuse of web search engines by plagiarists due to the exponential growth of internet data. In order to reduce the misuse of search engines, Liu et al. [24] identified suspicious documents. This is accomplished by working together with search engines to extract and validate content. The priority of retrieved text segments during plagiarism detection, which minimizes repetitive and pointless work, is a significant advance. In the context of plagiarism detection, this method successfully tackles the difficulties caused by data overload and improper usage of search engines. Although this approach is useful in decreasing search engine abuse, it might not fully address every aspect of plagiarism. The necessity for a comprehensive strategy is underscored by the difficulties associated with measuring textual overlap, online plagiarism, and paraphrase detection simultaneously.

While previous studies explore numerous plagiarism detection methods, it is clear that none of them fully handle some common forms of plagiarism, including simultaneous measurement of textual overlap, online plagiarism, and paraphrasing detection. This study's main contribution is the creation of a sophisticated method for detecting plagiarism that can concurrently spot several different types of academic dishonesty. Our proposed system has been developed to fully address the complex nature of plagiarism, encompassing not only traditional forms but also online plagiarism and textual overlap. In comparison to existing methods, which frequently focus on isolated aspects of plagiarism, such as direct copying or paraphrasing. This study significantly improves the effectiveness and adaptability of plagiarism detection techniques using a comprehensive approach, achieving the evolving demands of academic purposes.

3 Methodology

This section provides a detailed explanation of our approach for identifying direct copy, semantic changes, and online similarity in written content. We implemented a step-by-step procedure to ensure that these problems are identified correctly. Our procedure consists of three phases: data collection, textual similarity computation and identification, and online similarity detection.

Let's first discuss how our method identifies academic integrity. Here, the procedure of detecting academic integrity is shown on Algorithm 1. To detect academic integrity, the proposed methodology combines structural similarity (TF-IDF and cosine similarity) for detecting direct copy with semantic similarity (BERT embeddings) to identify semantic changes. Both methods are necessary, as structural similarity detects exact copying, while semantic similarity

identifies paraphrased or rephrased content. They collaborate by comparing similarity scores from TF-IDF (cosine) and BERT, using the higher score to classify plagiarism. This approach can identify verbatim copying, paraphrasing, and mixed cases, ensuring comprehensive detection. By taking the maximum similarity score from both analyses and comparing it to a predefined threshold (e.g., 30%), the methods identify direct copy and semantic changes. If the similarity is below 30%, check for online content; otherwise, it is copied from another student, either directly or with semantic changes. If the similarity score from cosine similarity is higher than the score from BERT, it indicates that the content has been copied directly. Otherwise, if the BERT similarity score is higher, it suggests that the content has undergone semantic changes or paraphrasing. If the similarity score is below 30%, the system further checks whether the content has been copied from online sources or not. If the content is not copied from online sources, then the content is original. By combining structural similarity (TF-IDF and cosine similarity) to identify lexical overlapping and semantic similarity (BERT embeddings) to capture contextual meaning, the study categorizes genuine content from paraphrasing. While paraphrasing shows low structural but high semantic similarity, genuine content shows low similarity on both measures.

After outlining the process for identifying academic integrity, this study moves on each step in more depth. Every stage is essential to this all-encompassing plagiarism detection approach. Examining these steps provides a comprehensive understanding of the methodology's importance, including data gathering, textual similarity calculation, similarity identification, and online plagiarism detection, all contributing to the integrity of written material.

3.1 First phase: data collection

The systematic collection of data from a wide range of sources within our academic department is the focus of the first stage of our proposed methodology. The system ensures data security and confidentiality by allowing students to upload assignments only through Google Forms or Google Classroom, with submissions directly stored in a Google Drive folder accessible exclusively to the course teacher. Students cannot view or track others' submissions, maintaining strict privacy. This setup ensures secure and confidential management of academic data while restricting access to authorized personnel only. Additionally, all data is stored on Google Drive, which employs encryption for data during transit, ensuring compliance with standard data security protocols. This crucial stage establishes the groundwork for

Algorithm 1 Algorithm for detecting Academic Integrity

```
pair_no ← 1
for i ← 1 to document.size() do
    for j ← i + 1 to document.size() do
        cosine_similarities[pair_no] ← cosine_similarities(document[i], document[j])
        bert_similarities[pair_no] ← bert_similarities(document[i], document[j])
        Similarities[pair_no] ← max(cosine_similarities[pair_no], bert_similarities[pair_no])
        pair[pair_no] ← (document[i], document[j])
        pair_no ← pair_no + 1
    end for
end for
total_pair ← nC2
Threshold_Online ← 30
for i ← 1 to total_pair do
    if Similarities[i] > Threshold_Textual then
        Plagiarized Content or Paraphrased Content
    else if Similarities[i] <= Threshold_Online then
        check_Online_similarities
    end if
end for
```

further analysis and the detection of plagiarism and paraphrase in academic documents.

3.1.1 Goal of data gathering

Creating a comprehensive and representative collection of materials written by students in our academic department is the main goal of this phase. In order to build a robust corpus that accurately represents the breadth of writing styles, topic matter, and academic levels widespread among our students, we strive to collect a wide variety of documents.

3.1.2 Data store

The data comprises a diverse range of academic documents (PDF, PPTX, DOCX, and TXT) submitted by students as part of their coursework and assignments. Here, if a student needs to resubmit an assignment, they must first remove the previously submitted document before submitting the new one. This process is managed automatically to ensure that only the latest version is retained, avoiding unnecessary comparisons of multiple versions of the same document. Additionally, there will be a specified submission deadline, after which all submitted documents will be converted into plain text files for uniform analysis. This approach ensures efficient handling of document resubmissions while maintaining an organized and streamlined evaluation process. These submissions span multiple topics and academic levels within a department, excluding programming assignments. The images and tables excluded from the processing. Given a set of n documents, all possible unique pairwise comparisons are computed, yielding $\binom{n}{2}$ combinations. The combinatorial formula [25] is presented on equation 1 which can be used to get the total number of pairwise document comparisons in a dataset with n documents:

$$\binom{n}{2} = {}^{n}C_2 = \frac{n!}{2!(n-2)!} \tag{1}$$

This ensures that each document is compared to every other document exactly once. Then the information is organized into a CSV file, which improves efficiency for future use. The process of obtaining and gathering data is depicted in Fig. 1.

To augment this collection, we search the archives and repositories of the department for historical texts and papers authored by previous students. Furthermore, we include documents authored by faculty, supervisors, and academic staff from our department, which greatly enhances the diversity of the dataset. Data is collected and stored in a designated folder on Google Drive. The data includes a total of 100 documents, with pairwise comparisons organized into a CSV file containing 4950 entries. Additionally, the methodology has been validated using a publicly available dataset the MIT Plagiarism Detection Dataset, accessible at https://www.kaggle.com/datasets/ruvelpereira/mit-plagiarism-detection-dataset#sourceKaggle [26] with a length of 367,375 samples, ensuring broader applicability and generalization of the results.

3.1.3 Data preprocessing for detecting plagiarism

To develop a plagiarism detection system, it's crucial to prepare data properly. This involves standardizing and optimizing input documents for comparison and similarity evaluation. The aim is to process raw text data to enhance its quality, eliminate unnecessary elements, and create

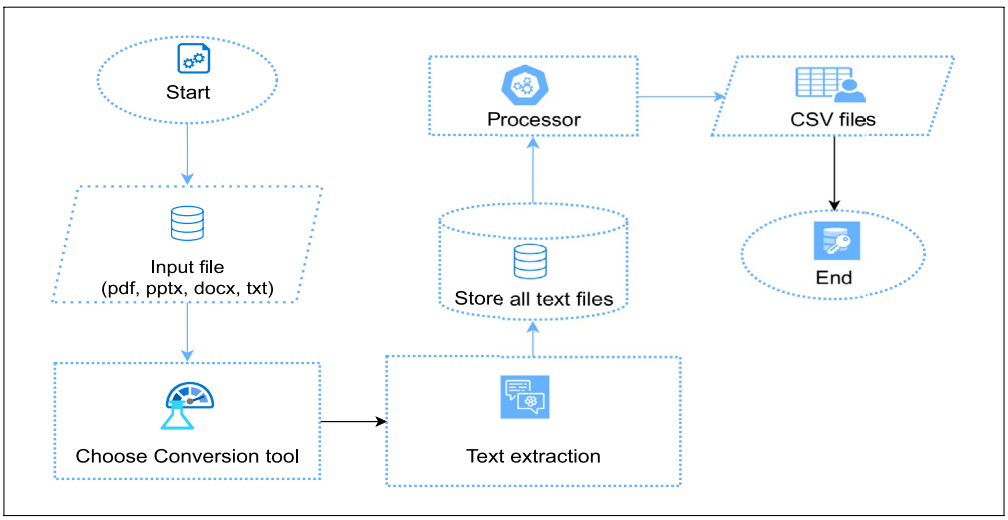


Fig. 1 Data collection and storage process for plagiarism detection

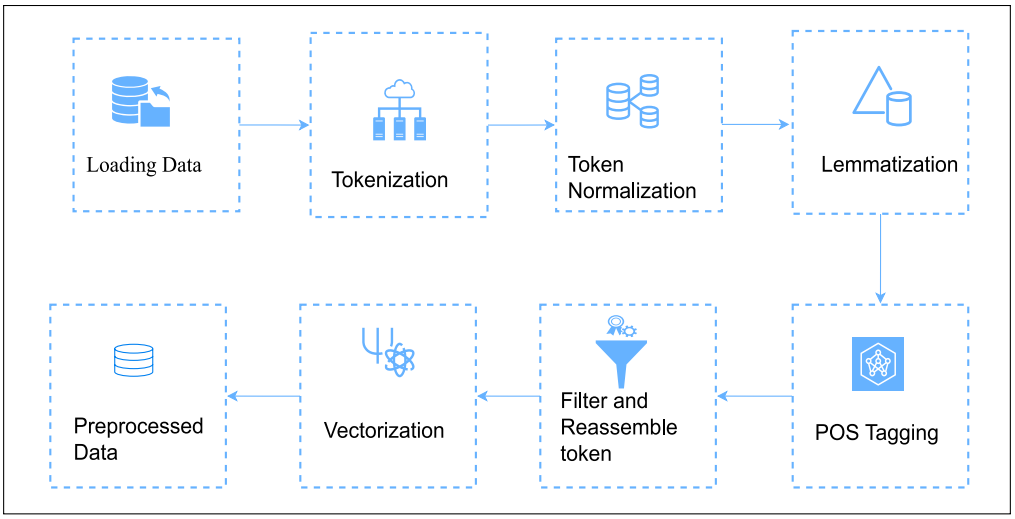


Fig. 2 Data Preprocessing Workflow for Textual Analysis

organized representations that can be used for similarity analysis. The ultimate goal of data preparation is to produce structured, high-quality textual data that is ready for further analysis.

The preprocessing step begins with loading the dataset containing the sentences to be processed. First, sentences are tokenized into individual words, and the tokens are cleaned by removing punctuation, stopwords, and non-alphabetic characters. This is followed by converting all tokens to lowercase. Next, lemmatization is applied to the tokens, reducing them to their base forms. After lemmatization, each token is assigned a part of speech (POS) tag, allowing for a more accurate treatment of words based on their grammatical role. The process continues with the removal of tokens that are not purely alphabetic or that are common stopwords, such as "and" or "the." Additionally, any tokens shorter than three characters are discarded. Finally, the remaining tokens are reassembled into a single preprocessed sentence, resulting in clean and standardized text ready for analysis. Then, the remaining tokens are reassembled into a single preprocessed sentence, resulting in clean and standardized text ready for analysis. Finally, the text is vectorized into numerical representations for analysis. Figure 2 presents an illustration of the data preparation method.

1. **Load Dataset:**
   - Load the dataset containing the sentences to be processed.
2. **Tokenization:**
   - Tokenize sentences into individual words using the tokenizer function:

     $$\text{Tokens} = \text{Tokenize}(\text{Sentences})$$

3. **Clean Tokens:**
   - Remove punctuation, stopwords, non-alphabetic characters, and convert tokens to lowercase:

     $$\text{Clean Tokens} = \{t \mid t \in \text{Tokens}, t \text{ is alphabetic and not a stopword}\}$$

4. **Eliminating Stop Words:** The initial step in data preprocessing involves cleaning the text [27], which includes the following essential tasks:
   - Unnecessary Character Removal: Unnecessary characters like punctuation, special symbols, and non-alphanumeric characters are removed from raw text data using the "re.sub" function and a regular expression pattern [28]. This leaves only words and spaces for analysis.
   - Lowercasing: Text is transformed to lowercase for case-insensitive similarity analysis. This standardizes the text and reduces matching complexity [29].

   Text cleaning produces a clean version of the input text that is prepared for subsequent processing.
5. **Lemmatization:**
   - Apply lemmatization to reduce tokens to their base forms:

     $$\text{Lemmatized Tokens} = \text{Lemmatize}(\text{Clean Tokens})$$

6. **POS Tagging:**
   - Assign Part-of-Speech (POS) tags to tokens for further lemmatization:

     $$\text{Tagged Tokens} = \text{POS_Tag}(\text{Lemmatized Tokens})$$

7. **Filter Tokens:**

- Remove non-alphabetic tokens, common stopwords, and tokens shorter than three characters:

$$\text{Filtered Tokens} = \{t \mid t \in \text{Tagged Tokens}, |t| > 2 \text{ and } t \text{ is alphabetic}\}$$

8. **Reassemble Tokens:**
   - Reassemble the remaining tokens into a single preprocessed sentence:

$$\text{Preprocessed Sentence} = \text{Join}(\text{Filtered Tokens})$$

9. **TF-IDF Vectorization:** A crucial text analysis approach known as TF-IDF vectorization converts the cleaned text data into numerical representations appropriate for quantitative research. The foundation of TF-IDF is the idea that the value and content of a text can be inferred from the frequency with which certain words or concepts appear in that document compared to other corpora. The following actions are part of the TF-IDF vectorization process:

   (a) Tokenization: The text has been cleaned and separated into its individual words and keywords. Every phrase is regarded as a sign.
   (b) Term Frequency (TF): The frequency of each term is computed for each document. In essence, this is a tally of the number of times each term appears in the text.
   (c) IDF (Inverse Document Frequency): This statistic gauges a term's uniqueness throughout the entire corpus. A higher IDF score is awarded to words that are uncommon throughout the corpus.
   (d) Calculating the TF-IDF score: involves multiplying the term TF by the IDF for each term in each document. Each document is then represented numerically, with each term contributing to the final score. The formula[30] for tf-idf is presented on equation 2:

$$W_{y,j} = \text{TF}_{y,j} + \ln\left(\frac{N}{\text{df}_y}\right) \tag{2}$$

Each document is represented as a vector of TF-IDF scores for its constituent terms in the TF-IDF matrix, which is a structured representation of the documents in the corpus. In later stages, such as similarity analysis, this matrix is utilized to calculate how similar two documents are based on the phrases they share.

In order to detect plagiarism, it is crucial to clean up the text by removing unwanted elements and standardizing the format, which is known as data preprocessing. Here, lemmatization and POS tagging make the text more consistent by reducing words to their base forms and identifying their roles, helping the system better understand and compare the content. These techniques ensure that the system focuses on the meaning of the text, reducing noise and redundancy. While they are more effective for structural analysis and similarity detection, they are complemented by semantic methods like BERT embeddings, which capture more contextual analysis. Together, these techniques provide a robust foundation for analyzing and comparing textual content. After that, TF-IDF vectorization is used to transform the text into numerical representations. These processed representations serve as the basis for plagiarism detection algorithms and are crucial for analyzing similarities. The dataset created during this phase is essential for detecting plagiarism and paraphrasing in later stages.

3.1.4 Training and testing model

In our study, the dataset was partitioned into training and testing subsets, with 80% of the data allocated for training purposes and 20% reserved for testing. This approach ensures a robust evaluation of the model's performance. The study also utilized 10% MIT Plagiarism Detection Dataset for testing, which is publicly available and can be accessed

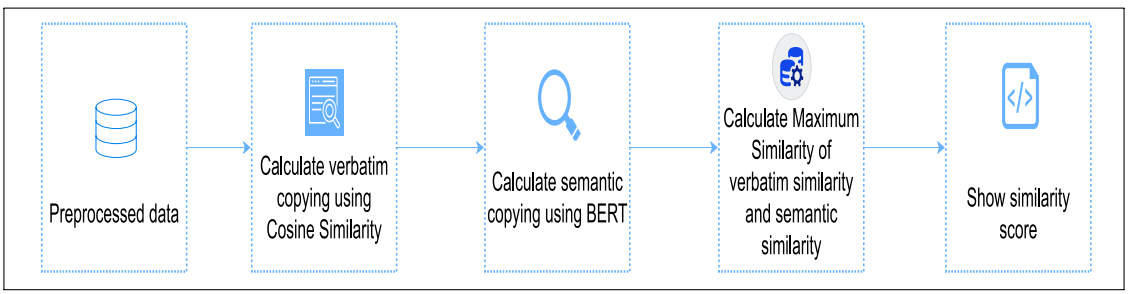


Fig. 3 Similarity detection process

at the following link: https://www.kaggle.com/datasets/ruvelpereira/mit-plagairism-detection-dataset#sourcedata link.

3.2 Second phase: detection of similarities

In our methodology, we incorporate a process named similarity detection to determine similarity between academic papers. This step includes a comprehensive analysis of the text's substance and contextual details to determine cases of plagiarism or rewording[31]. We utilize two primary methods for evaluating similarity: Cosine Similarity and BERT-based paraphrase identification. Then the maximum of these two similarities used for further processing. The detailed procedure for similarity detection in the methodology is shown in Fig. 3.

1. Cosine Similarity: A popular metric for determining how similar two vectors are in a high-dimensional space is cosine similarity. The cosine similarity [32] formula is presented on equation 3:

$$\text{Cosine Similarity}(S_1, S_2) = \frac{\sum_{i=1}^{n} S_1(i) \times S_2(i)}{\sqrt{\sum_{i=1}^{n} S_1(i)^2} \times \sqrt{\sum_{i=1}^{n} S_2(i)^2}} \quad (3)$$

   These vectors represent the TF-IDF representations of academic documents in the context of our research. The relevance of each term within the document and across the entire corpus is captured by the TF-IDF matrix, which is built by converting the text data into numerical characteristics. This numerical representation enables the identification of terms that are both significant within individual documents (TF) and unique across the corpus (IDF). By weighting these terms appropriately, TF-IDF highlights key features that contribute to assessing the relevance and originality of documents in comparison to others discussed in this article [33]. The cosine of the angle between any two TF-IDF vectors is used to calculate their similarity. This measure has a value range of -1 to 1, is converted to a percentage by scaling the value to fall within a range of 0% to 100%, where higher percentages indicate greater similarity between the compared vectors. It aids in the identification of documents that have comparable word usage patterns and structures when applied to our dataset.
2. BERT based Semantic Changes Detection: We use cutting-edge Natural Language Processing (NLP) [34] methods, notably combining Sentence Transformer [35] and BERT model, in addition to Cosine Similarity, for semantic changes detection. The Sentence Transformer generates dense embeddings for sentences, optimized to capture semantic similarity. Using the all-MiniLM-L6-v2 model [36], we compute semantic similarity scores with cosine similarity, ensuring efficient and robust analysis. Then, we load a pre-trained BERT model and tokenizer for paraphrase text identification. BERT's tokenizer is used to tokenize and encode each pair of sentences. The BERT model processes the encoded inputs and calculates the paraphrase logits. After that, we use a softmax method to get a percentage-based paraphrase similarity score. This number aids in determining the likelihood that two sentences are paraphrases. We employed the pre-trained bert-base-uncased model from Hugging Face, which is intended for sequence classification with two labels, to fine-tune BERT. The model was trained with the BertTokenizer to allow tokenization, and threshold improving was used to achieve the best paraphrase detection performance. The model was optimized using the AdamW optimizer with a learning rate of 1×10^{-5}.

Using the maximum of BERT-based and cosine similarity scores combines their complementary strengths: BERT captures semantic nuances while cosine similarity evaluates term frequency patterns. This approach ensures a more comprehensive and accurate measure of text similarity, enhancing the detection of potential plagiarism or rewording. Then we further refine our analysis based on the observed similarity scores in the following phases of our technique. In order to determine the level of textual overlap, this study checks if it is plagiarized content or paraphrased content if the similarity score is greater than the threshold 30%. If the plagiarism similarity is higher, the content is identified as plagiarized. If the paraphrasing similarity exceeds the plagiarism similarity, the content is classified as paraphrased. Conversely, if the plagiarism similarity is higher, the content is identified as plagiarized. On the other hand, the online plagiarism detection phase proceeds to investigate possible outside sources if the similarity score is less than or equal to threshold 30%. Here, a threshold of less than 30% is used to identify cases for online plagiarism detection when similarity is lower but still important.

3.3 Third phase: online similarity detection

The online similarity detection phase begins if the similarity score from the second phase is less than or equal to 30%. The proposed method carefully examines the input text line by line to ensure thorough detection. For each line, an online search is conducted to find multiple relevant sources, and cosine similarity is calculated to compare the text against the content from these sources. Lines with a similarity score above a set threshold (e.g., 65%) are marked as "Copied," and all contributing sources are documented. The system then aggregates these findings

to calculate the overall plagiarism percentage, providing clear and detailed insights into the sources and the degree of similarity. This method ensures a comprehensive analysis for detecting plagiarism in multiple online sources.

The main objective is to distinguish between cases of low similarity due to potential paraphrasing or the content that has probably sourced from online resources. Low-similarity documents are carefully examined to determine whether they were obtained from online sources or not. The advantage of this phase is that it can provide a more thorough and sophisticated understanding of possible cases of online plagiarism, increasing the overall efficacy of the process in detecting plagiarism.

The Algorithm 2 explains how online plagiarism detection works by outlining the step-by-step process of examining text to identify potential similarities with online sources.It reads the text first, then searches the internet for anything comparable. In the event that nothing noteworthy is discovered, “No relevant online source found” appears. However, it retrieves the web information and determines a similarity score if it finds commonalities. The message “This text seems copied from an online source” is flagged if the score is high. “This text is unique and not copied online”, it promises in the event that the score is low. The purpose of this algorithm is to ensure that the writing is authentic and to detect instances of online plagiarism. Here we consider the threshold of 65% as we check when it is greater than or equal to 65%, then the document is mostly similar to online content.

Algorithm 2 Determine the Internet’s Text Similarity

```
file_text ← read_file(file_path)
online_urls ← generate_online_urls(file_text)
Threshold ← 65
if online_urls is empty then
    return "No relevant online source found."
end if
total_copy ← 0
total_lines ← count_lines(file_text)
for each line in file_text do
    for each url in online_urls do
        online_text ← scrape_online_content(url)
        if online_text is None then
            continue
        end if
        similarity_score ← calculate_similarity(line, online_text)
        if similarity_score ≥ Threshold then
            total_copy ← total_copy + 1
            break
        end if
    end for
end for
plagiarism_percentage ← (total_copy/total_lines) × 100
if plagiarism_percentage ≥ Threshold then
    return "This text appears to be copied from an online source."
else
    return "This text is unique and does not appear to be copied from an online source."
end if
```

3.3.1 Online content retrieval and data extraction

To get access to relevant text content from the web, we start by sending HTTP requests to online sources to fetch web content. The "Beautiful Soup (bs4)" library makes use of web scraping [37] techniques to enable us to extract useful data from the retrieved online information. We may explore document elements, parse HTML or XML documents, and collect useful text material with this potent tool.

3.3.2 Text vectorization and cosine similarity

The "TfidfVectorizer" class, which is a key component of the "Scikit-learn (sklearn)" library[38], is used in our search for accurate text comparison. The TF-IDF technique is used in this class to transform textual data into numerical representations, ensuring that the retrieved web material and the low similarity document may be analyzed together. Additionally, the scikit-learn "cosine-similarity" function quantifies the cosine similarity across vectors, allowing us to judge how similar the low similarity document and the internet material are.

3.3.3 Data integrity and text preprocessing

Our objective is to ensure data integrity, which involves the utilization of regular expressions (re) for precise text analysis and pattern matching [39]. Here, we use text preprocessing and pattern matching to remove extraneous text from the text data. This preliminary stage prepares the text for in-depth examination, ensuring the accuracy and dependability of our findings.

3.3.4 Effective online content retrieval

We use the "Googlesearch" package to speed up the process of finding online content. It efficiently retrieves relevant search results and simplifies the generation of search queries for documents with low similarity.

The outcomes of this step offer a precise evaluation of the user's text in comparison to internet sources. It assists in differentiating between various levels of plagiarism so that educators and institutions can take the proper action in accordance with the seriousness of the plagiarism that has been found. This study ensures a thorough and effective approach to plagiarism detection and analysis by

```
Index 241:

K-means Clustering Numerical Problem: Apply K -means clustering algorithm for
the follo wing dataset. X0 Y0 185 72 170 56
168 60 179 68 182 72 188 77 Solution: Let,
c1 = (185,72) & c2 = (170,56) Iteration 1: (185,72)
(170,56) (168,60) (179,68) (182,72) (188,77) Distance from c1 = (185,72)
0 21.93 20.81 7.21 3 5.83 Distance from c2 =
(170,56) 21.93 0 4.47 15 20 27.66 Assign group C1
```

Fig. 4 Text data before pre-processing and segmentation

```
Index 241:
clustering numerical problem apply k clustering algorithm follo
wing dataset solution let iteration distance distance assign
group new centroid x x iteration distance distance
assign group new centroid x x an application
clustering used variety example business case real life
like academic performance diagnostic system search engine wireless
```

Fig. 5 Text data after pre-processing and segmentation

```
File: 18CSE031.pdf-18CSE039.pdf, Similarity Percentage = 70.98%
File: 18CSE031.pdf-18CSE024.docx, Similarity Percentage = 70.96%
File: 18CSE031.pdf-18CSE029.pdf, Similarity Percentage = 70.48%
File: 18CSE031.pdf-18CSE010 .pdf, Similarity Percentage = 71.02%
File: 18CSE031.pdf-18CSE037.pdf, Similarity Percentage = 70.99%
File: 18CSE031.pdf-18CSE034.pdf, Similarity Percentage = 65.71%
File: 18CSE031.pdf-18CSE006.docx, Similarity Percentage = 71.17%
File: 18CSE031.pdf-18CSE004.pdf, Similarity Percentage = 70.96%
File: 18CSE031.pdf-18CSE044.pdf, Similarity Percentage = 71.62%
File: 18CSE031.pdf-17CSE002.pdf, Similarity Percentage = 70.59%
```

Fig. 6 Cumulative similarity percentage of student submissions

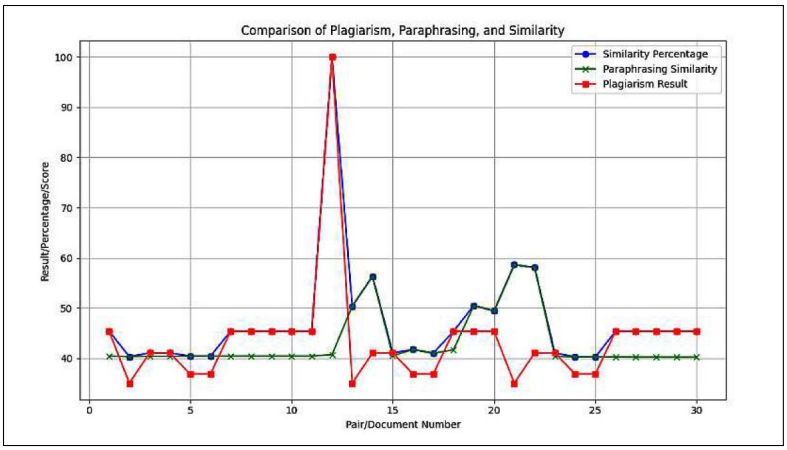


Fig. 7 Line chart for the comparison of plagiarism using plagiarism result, paraphrasing similarities and similarity percentage

methodically carrying out these processes, helping you maintain originality and academic integrity.

Our multi-phase process offers a reliable and adaptable solution for paraphrase analysis and plagiarism detection. We seek to guarantee the originality and authenticity of academic documents by integrating online cross-referencing with structural and semantic similarity analyses. This approach provides educators, organizations, and students with a potent instrument to protect academic integrity and advance an ethical and rigorous intellectual culture.

4 Results and discussion

We demonstrate the outcomes of our multi-phase plagiarism detection technology in this part. In the context of evaluating document similarity and maintaining academic integrity, we talk about the results and their significance.

The HP Probook, Intel Core i5 and 2.80 GHz CPU, 8GB RAM, Windows 10 Pro 64-bit, and Python 3.10.12 were used to apply the steps of the proposed application to a number of suspicious documents. Any sort of document, whether a pdf, doc, or docx, must be converted to text in order to be used. The text file folder will then be converted to a csv file.

Figure 4 illustrates a sample of the raw data before any preprocessing has been conducted. The figure depicts one index of row data. The data is displayed as it appears in its original form, allowing for an assessment of its initial state before any modifications or enhancements are made.

To assure the accuracy and dependability of our findings, we carried out necessary data pretreatment procedures before to completing the various rounds of our plagiarism detection methodology.

The text pre-processing results are illustrated in Fig. 5 along with the segmentation of the pre-processed text data into words for text analysis tasks like text classification or natural language processing.

To evaluate the structural and semantic similarities of academic documents, we used Cosine Similarity and a BERT-based Paraphrase Detection model in the second stage of our research. The integrated method offers a thorough comprehension of plagiarism. The cumulative similarity percentage obtained from the phase is shown in Fig. 6. The files include student submissions labeled as, for instance,

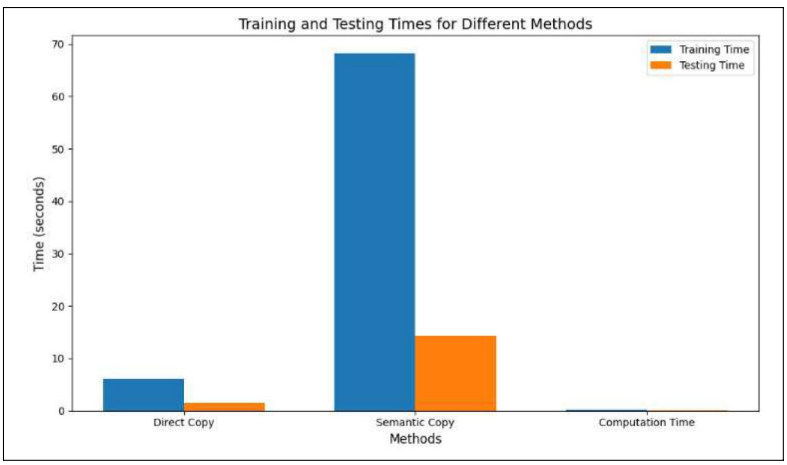


Fig. 8 Different plagiarism detector models' times of execution

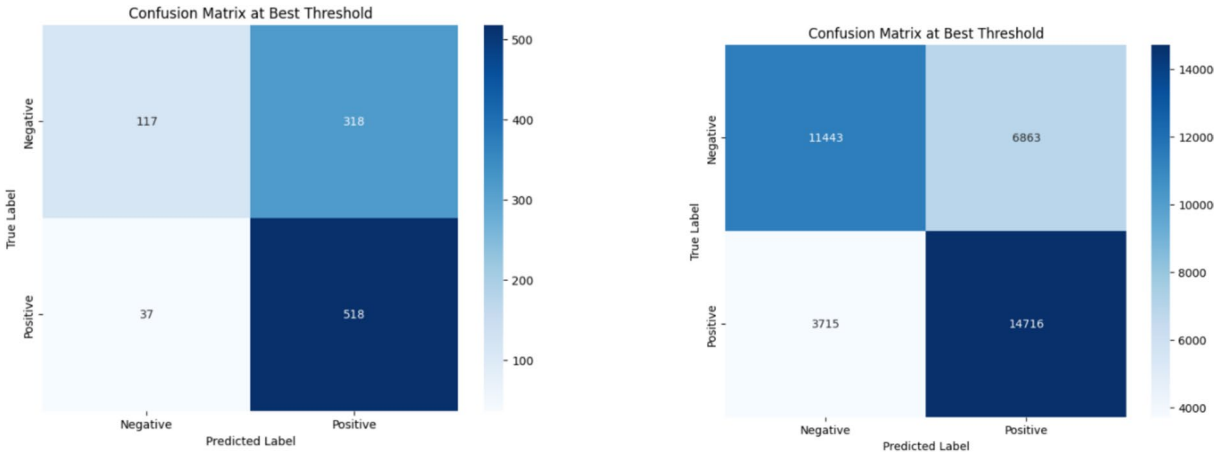


Fig. 9 Confusion matrices of similarity detection phase

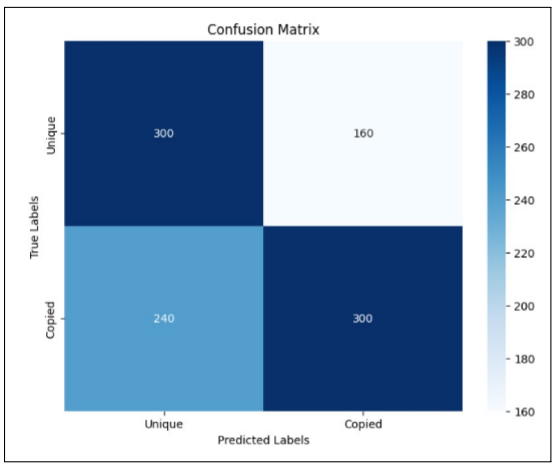


Fig. 10 Confusion matrix of online similarity detection method

"18CSE031.pdf" and "18CSE039.pdf", which were turned in for evaluation. The given similarity percentage is intended to measure how similar these two particular texts are to one another.

Higher values denote a stronger similarity in document structure. The Cosine Similarity values, which range from 0 to 1, describe the structural similarity of texts. The Paraphrase Score values, which likewise range from 0 to 1, measure the degree of paraphrastic link between documents; higher scores suggest greater semantic similarity. Three metrics (plagiarism outcomes, paraphrasing similarities, and similarity percentages) are represented visually in Fig. 7 for various pairs of textual data. The figure shows the computed similarity score is the greatest value obtained from the BERT model and cosine similarity. This combination of methods allows for precise determination of the

Table 1 Comparison of plagiarism detection approaches

References	Methodology	Coverage	Key ccontributions
Chang et al. [40]	Word2vec	Text documents	False Discovery Rate (FDR) of 2.56%, an Accuracy Rate (AR) of 72.91%, and a Completion Rate (CR) of 74.83%
Ratna et al. [41]	Linear vector quantization	Automated cross-language plagiarism detection	Computerized cross-language plagiarism detector with accuracy: 65.98%
Veisi et al. [42]	Sentence vectors, word vectors with Levenshtein distance	Detect plagiarism in Persian texts	Text alignment plagiarism detection with plagdet of 95.87%
Goyal et al. [43]	Exact string matching, cosine similarity, and database/web searches	Web-based plagiarism detection system	Accuracy for sets P (89.79%), Q (73.58%), R (86.95%), S (48.3%), U (96.07%), and V (61.53%)
Arabi et al. [44]	Semantic vector formation with FastText and WordNet, structural matrix weighting with TF-IDF	Research papers	Detects extrinsic plagiarism with 95.1% precision for one method and 93.8% precision, for another
This Study	Cosine Similarity, BERT-based Paraphrase Detection, Web Scraping	Academic documents considering both structural and semantic similarities	Comprehensive approach, with an f1-score of 74% for semantic similarity detection and 60% for online similarity detection

similarity percentage between two documents, which helps identify paraphrase instances. If the similarity percentage matches the paraphrasing similarity, the content is identified as semantically similar, whereas if it matches the plagiarism outcome, the content is identified as a direct copy.

The execution time comparisons for different models are calculated using our private dataset evaluated using a standard computational environment (HP Probook, Intel Core i5, 8GB RAM) and a dataset representative of practical use cases. The execution times for tasks like plagiarism detection and paraphrasing similarity reflect the method's efficiency in handling real-world use cases, demonstrating its ability to balance efficiency with precision for academic applications. The training execution times for Direct Copy, Semantic Copy, and the time for calculating the maximum time between direct copy and semantic copy are 6.11, 68.21, and 0.00396 s, respectively. The testing execution times for Direct Copy, Semantic Copy, and the time for calculating the maximum time between direct copy and semantic copy are 1.55, 14.25, and 0.00099 s, respectively. Here, Fig. 8 displays both the training and testing time. The Semantic Copy method takes longer time than the others.

A threshold of 0.3 yielded the best performance in similarity detection. For our private dataset our model achieved an average precision of 62% and an average recall of 93%. And the error rate, accuracy, f1-score for this threshold are 36%, 64%, 74% for the private dataset. For MIT plagiarsm dataset our model achieved an average precision of 68% and an average recall of 80%. And the error rate, accuracy, f1-score for this threshold are 29%, 71%, 74% for MIT Plagiarism dataset. The assessment measures offer a reliable framework for evaluating the efficacy of the proposed method, as they thoroughly represent its effectiveness in balancing essential performance metrics such as accuracy, precision, and recall in plagiarism detection. Figure 9a illustrates the confusion matrix for the private dataset, and Fig. 9b illustrates the confusion matrix for the MIT Plagiarism dataset of the similarity detection phase.

The confusion matrix analysis provides valuable insights into the system's strengths and areas for improvement. For our private dataset, the model shows excellent recall at 93%, along with a balanced F1 score of 74%, indicating high performance in detecting positive cases. However, moderate precision (62%) and accuracy (64%) indicate the need to further reduce false positives and improve overall classification accuracy. For the publicly available MIT Plagiarism Detection Dataset, the model achieves a strong recall of 79% and a balanced F1 score of 74%, effectively identifying 14,716 true positives. However, a precision of 62% and 6,863 false positives indicate opportunities for improving the system. Testing on datasets of varying sizes demonstrates the system's adaptability to both small-scale applications and large-scale implementations, highlighting its flexibility in handling diverse data scenarios effectively. These findings illustrate that our method is robust, scalable, and generalized across datasets of varying size and complexity. The model's high recall across both datasets ensures reliable detection of plagiarized content, while its balanced F1 score indicates effective handling of diverse cases.

The online similarity method not only generates similarity results, but it also identifies the sources from which the student may have copied the content. The online similarity detection model achieved a precision of 65%. However, it has a recall of 56%, an accuracy of 60%, and an F1 score of 60%. Figure 10 depicts the confusion matrix of the online similarity detection method.

A comparative analysis of different approaches to plagiarism detection is given in Table 1. A comparative comparison of the current study and previous works in the field of plagiarism detection highlights a number of distinctive contributions, strengths, and novel aspects that together highlight the importance of the suggested methodology.

This study contributes significantly to the field of plagiarism detection by offering a flexible method that can handle many types of academic misconduct in a thorough manner. This technique, in contrast to conventional approaches, is highly effective in detecting instances of direct content copying, even in the presence of minor modifications like word substitutions. It offers a sophisticated classification of plagiarism into three distinct categories: direct copying, semantic change, and copying from online sources. Notably, the study addresses the issues brought on by the ease of access to information on the internet by introducing a novel feature that makes it possible to identify content that is directly drawn from online platforms. Furthermore, the methodology provides a numerical indicator to indicate the degree of content sourced from the internet, improving transparency when assessing reliance on web-based content. Overall, these unique contributions position this approach as a valuable and comprehensive tool for effective plagiarism detection in academic documents.

The approach works well for handling problems with semantic changes because it makes use of BERT's capacity to comprehend natural language and context, which improves its ability to detect any changes in academic writing. Additionally, the methodology is strengthened by the integration of Web Scraping and Google Search API, which makes it possible to identify plagiarism that originates online and demonstrates a forward-thinking approach in the era of easily accessible online content.

The study is novel for its ability to detect different types of plagiarism, classify them into categories, and measure how much content is taken from the internet. After incorporated, these features make it a reliable and complete tool for identifying plagiarism in academic settings.

5 Conclusion and future direction

This study introduces a comprehensive approach for identifying and classifying plagiarism in academic contexts. Recognizing plagiarism as a persistent challenge in education, the study aims to offer a deeper understanding of its various forms. Using a multi-phase approach, it assesses a wide range of text similarity scores, emphasizing the need for a flexible and adaptive method to address diverse plagiarism scenarios. The study categorizes plagiarism into three distinct types: direct copy, semantic changes, and online copy, allowing for a more precise evaluation of textual overlap. This classification provides educators and institutions with valuable insights into the nature of potential misconduct, enabling specific responses and interventions. By incorporating online plagiarism detection, the study extends above the limits of internal document analysis. This step underscores the importance of addressing external sources of plagiarism and highlights the need for prevention strategies. Finally, the study establishes a robust framework for detecting and classifying plagiarism, integrating internal and external detection methods while offering a sophisticated understanding of similarity levels. Enhancing plagiarism detection techniques and developing preventive measures remain essential for preserving academic integrity in the digital age. This study contributes to ensuring the authenticity and originality of academic work.

In the future, we will improve the system's ability to improve online plagiarism detection, which is the capacity to identify instances of plagiarism where individuals copy content from online sources and then paraphrase it without giving credit. To address challenges like limited access to private or region-specific content and the changing nature of online material, future efforts will integrate local databases with real-time online searches. Additionally, incorporating larger and more diverse datasets will help refine and validate the system. Enhancements such as better feature selection, optimized thresholds, and advanced contextual embeddings will improve accuracy and scalability, ensuring the system remains reliable and effective in detecting plagiarism across various contexts.

Author Contributions Regarding author contributions, each author played a specific role in the paper. Deblina Mazumder Setu: Conceptualization, Methodology, Writing - original draft. Tania Islam: Validation, Supervision. Md Erfan: Incorporating Feedback. Samrat Kumar Dey: Reviewing manuscript, Supervision. Md.Rashid Al Asif: Monitoring Work Progress. Md Samsuddoha: Monitoring Work Progress

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Data Availability Due to confidentiality of student information, data is not available. For testing purposes, we also utilized a publicly available data set, which can be accessed at the following data link: https://www.kaggle.com/datasets/ruvelpereira/mit-plagairism-detection-dataset#sourcedata link.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This research strictly adheres to ethical guidelines. Permissions were obtained for the collection and use of data from the student participants. The study prioritizes the well-being and rights of the participants.

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